COMP 3610: Big Data Analytics

Personalisation Engine

Group 9: Taariq Motiram, Shivan Kissoon, David Williams, Danielle Gonzales

PROBLEM

What Problem Are We Solving?

How do we show the right content to the right user at the right time?

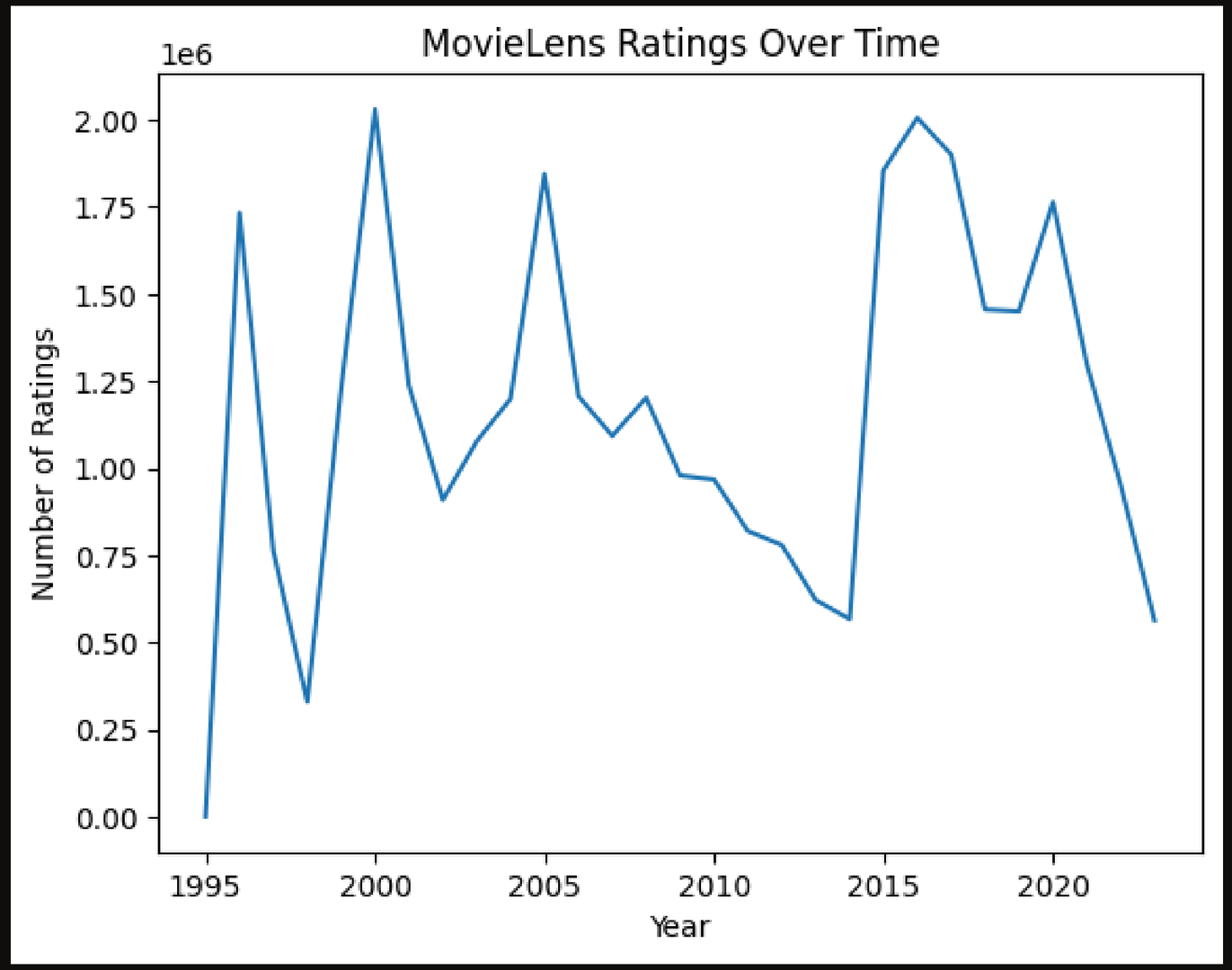
Cold-Start Problem

New users receive no meaningful recommendations due to lack of previous interactions

Goal: Build a personalisation engine that recommends items (movies) a user is most likely to click or interact with

DATA

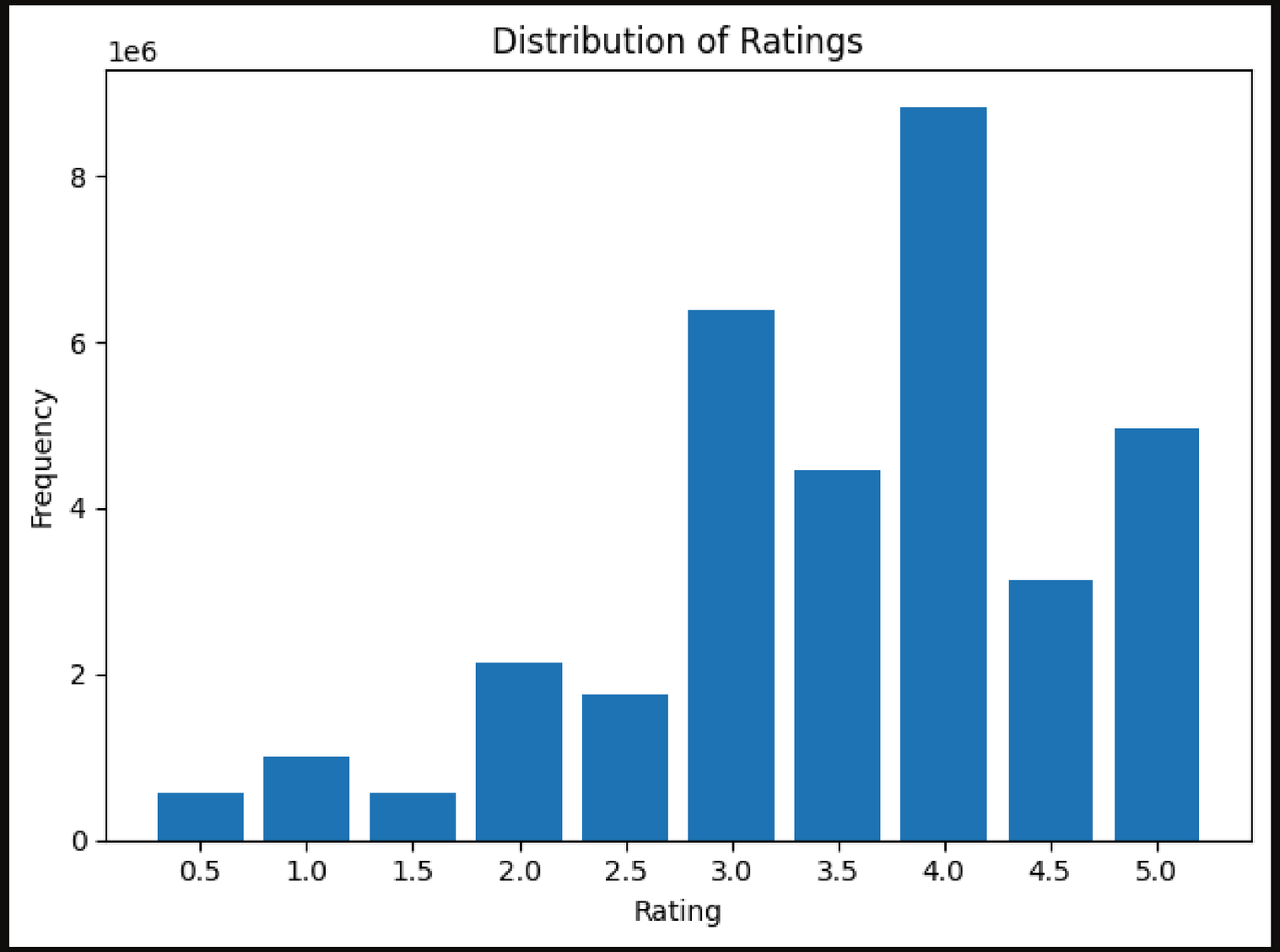
Dataset Overview



- 31 years of movie ratings
- The dataset was well mantained with zero missing values
- major peaks in early 2000s and 201t4s
- Drops in late 1990s, 2006 and early 2020s
- Total of 33,832,162 ratings
- Total of 86,537 movies

DATA

Dataset Overview



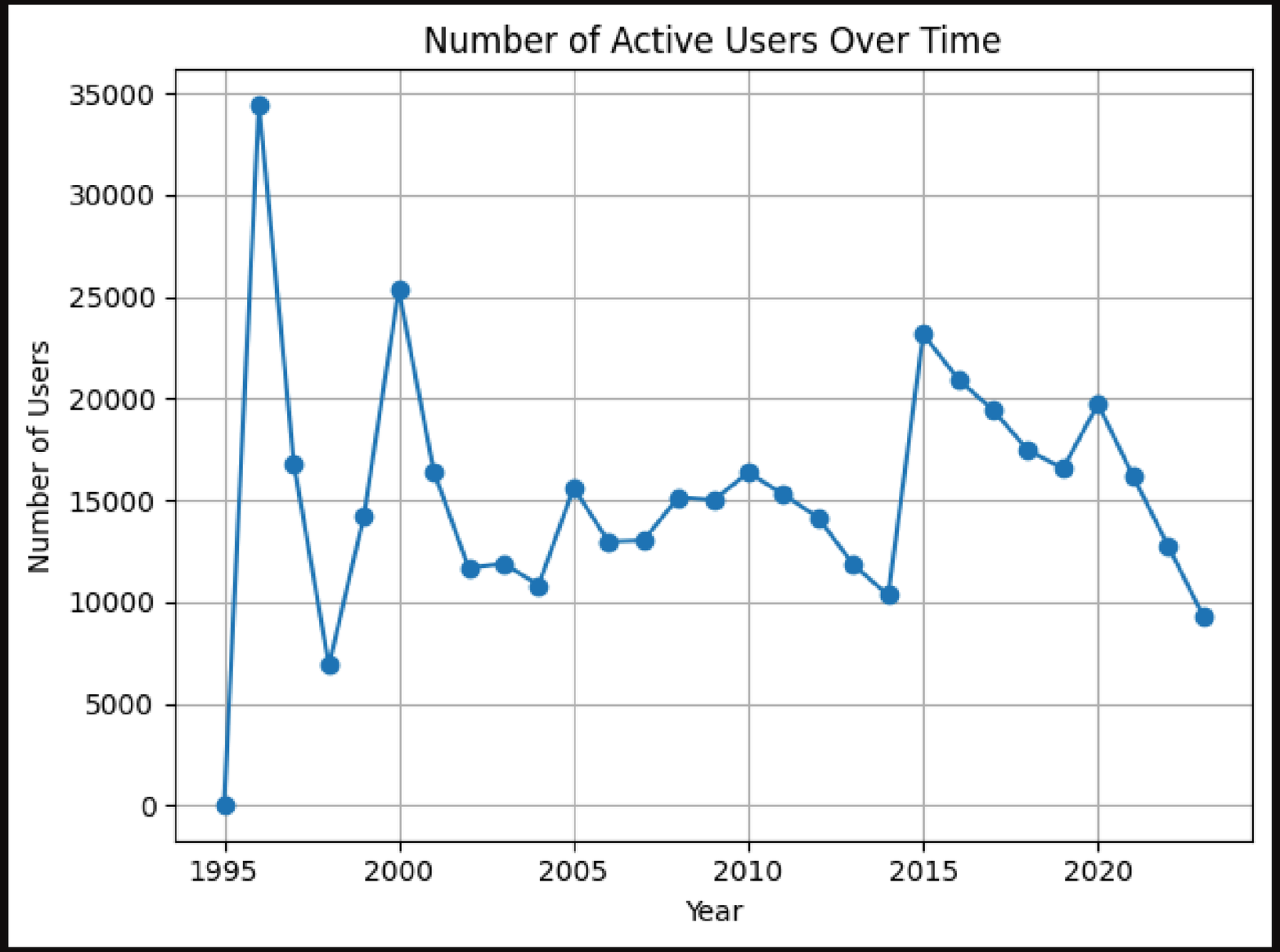
Average rating per movie: 3.5

Average number of rating per user: 3.7

Average number of rating per user was 102

Average number of rating per movie: 406

DATA



Peak users was in the late 1990s with 35,000

Minor peak in 2018 due to covid-19 pandemic

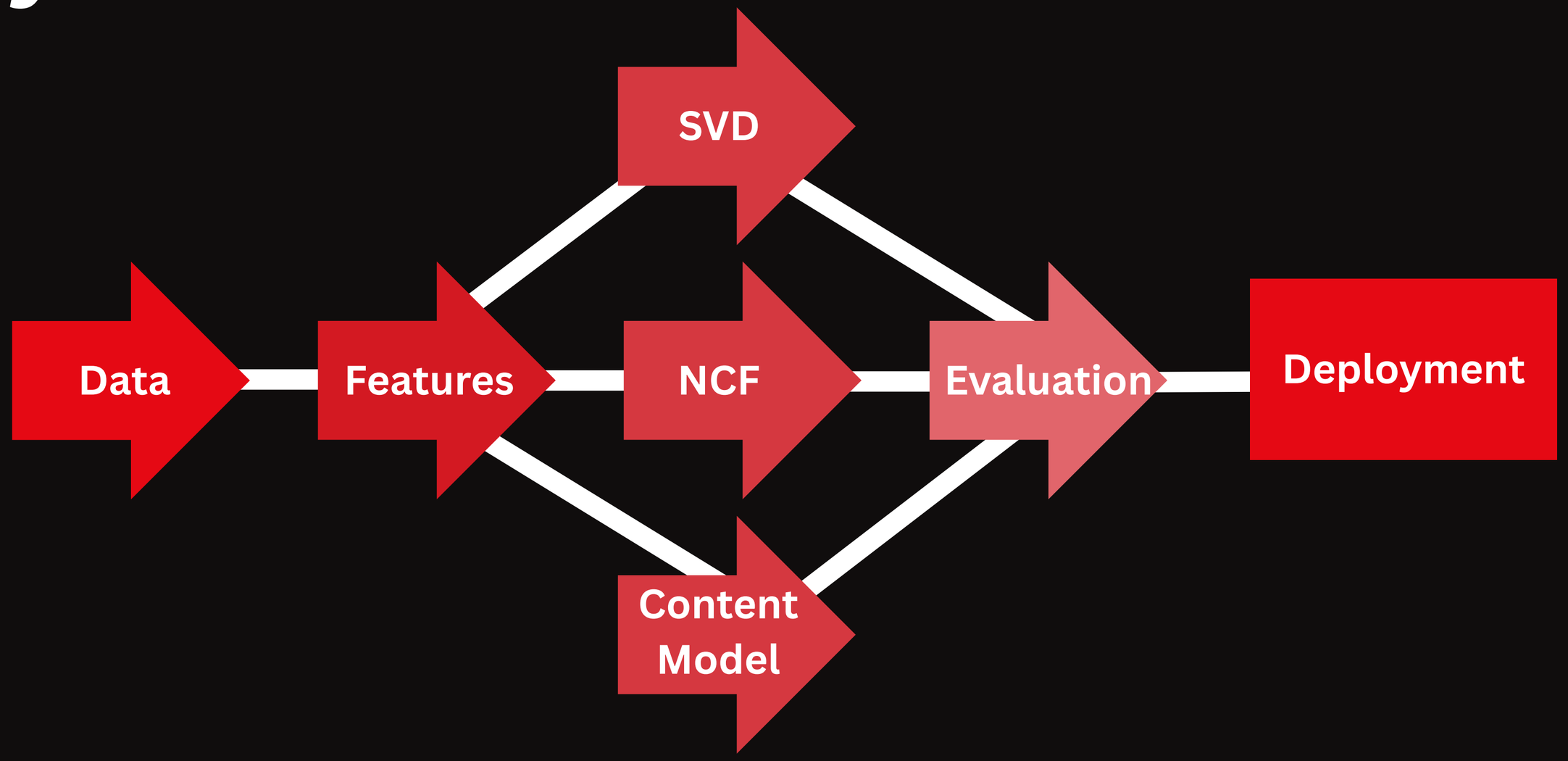
The most rated movie was The Man Who Came to Dinner with 180,585 ratings with avg of 5.0

The least rated movie was Besotted with 5805 ratings and avg of 0.5

METHODOLOGY

SVD

System Architecture



METHODOLOGY

Algorithm Choices + Justification

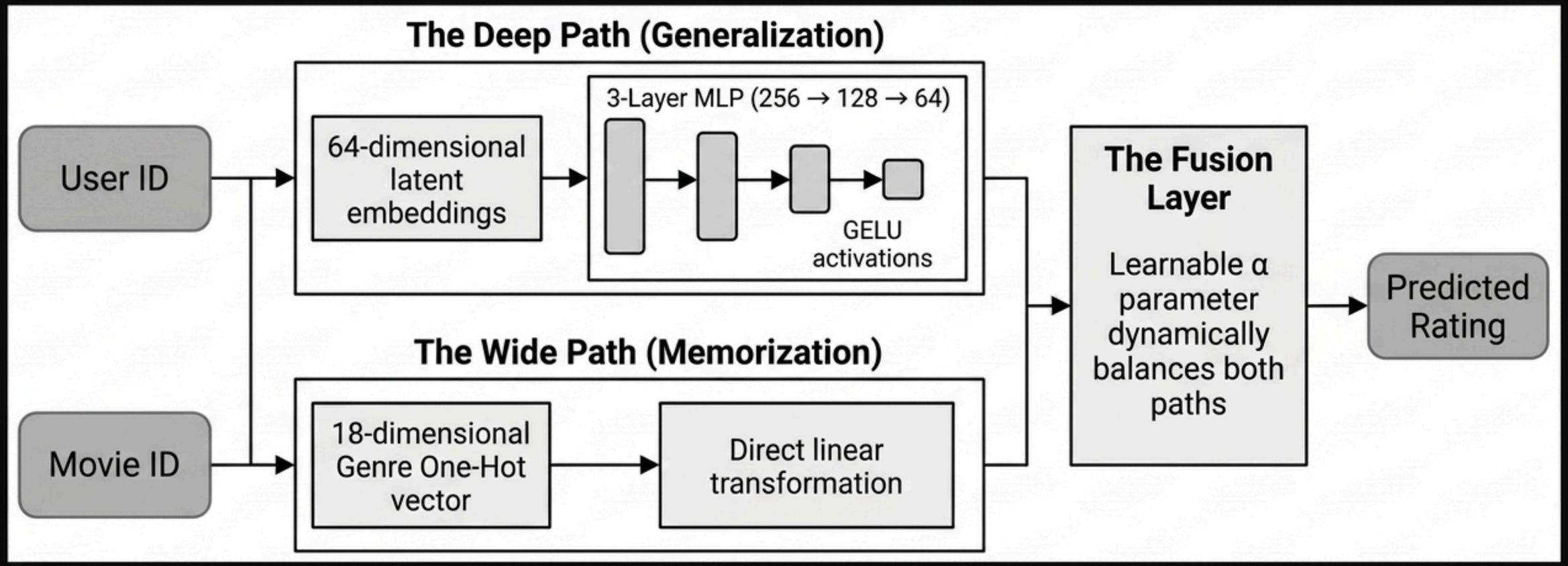
Singular Value Decomposition (SVD)

Neural Collaborative Filtering (NCF)

Content-Based Filtering (Genre)

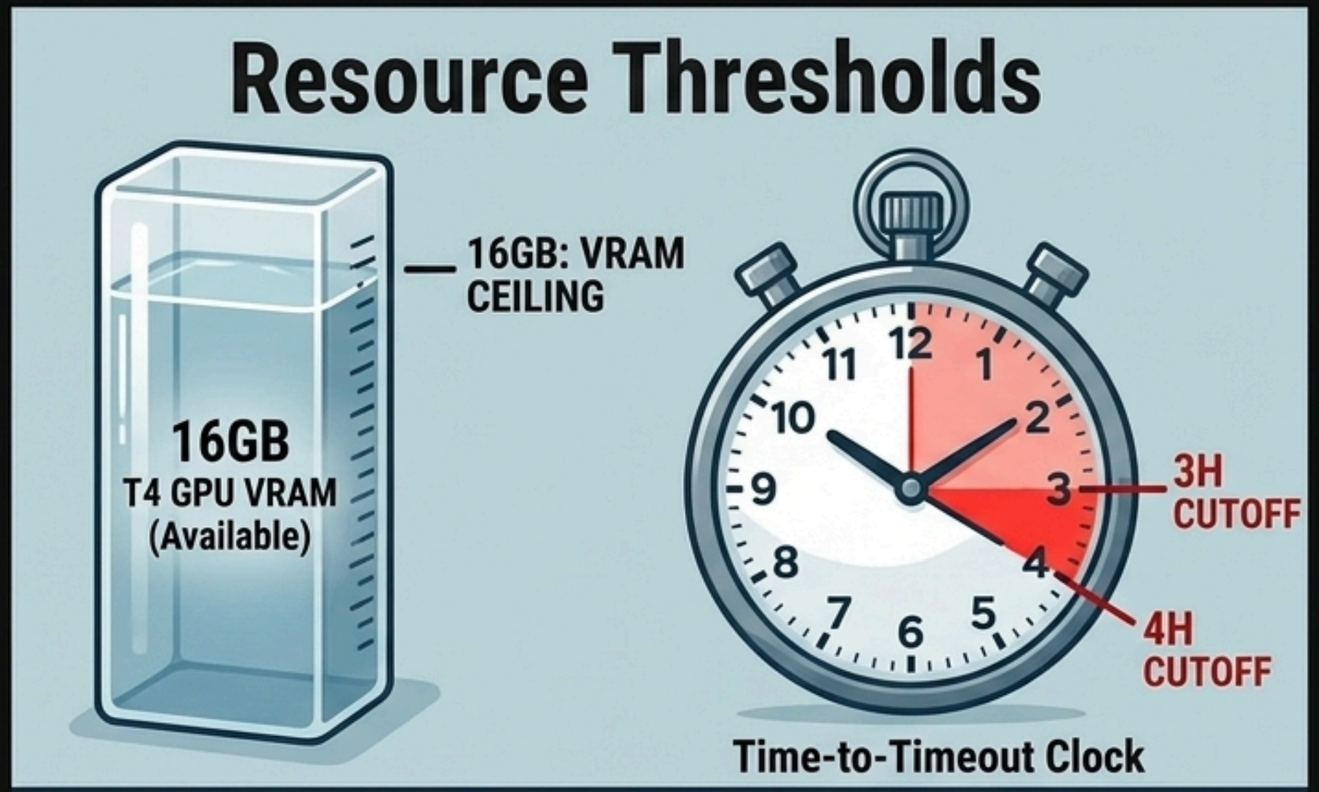
SYSTEM ARCHITECTURE

The Dual-Stream Approach



Deep Path: 64-dim embeddings, 3-Layer MLP (256 → 128 → 64), GELU |
Wide Path: 18-dim Genre Vector, linear transform | **Learnable** α balance

RESOURCE CONSTRAINTS & SCALING BOTTLENECKS



Hardware Limitations vs. Data Scale:
Project bound by Google Colab Free Tier (NVIDIA T4). Large-scale operations for 10M/25M datasets hit Memory Wall (16GB VRAM) during initialization.

Runtime Timeout:
MovieLens 1M converges in ~45 min. Runs on 10M/25M hits 3-hour kernel timeout before convergence.

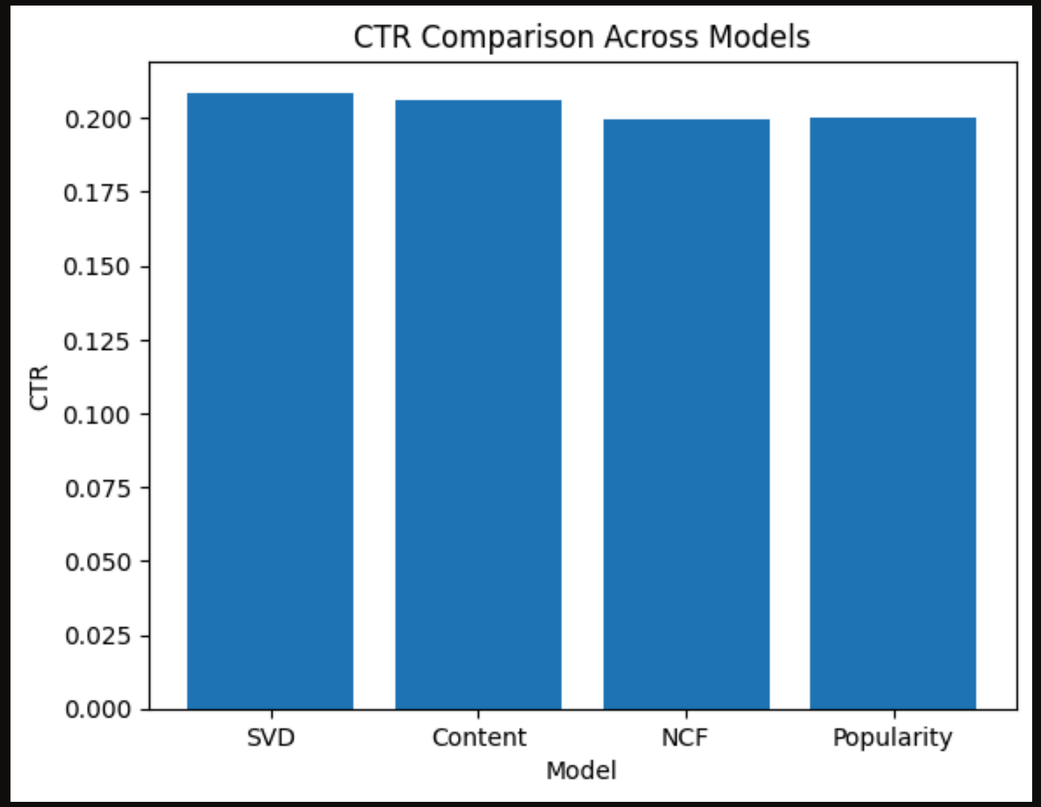
The Engineering Decision

- Prioritized Model Complexity over Data Volume.
- **Decision**: Stick to 1M dataset to implement advanced features (Attention Scorer, Wide & Deep Fusion).
- Avoided 'dumb' model trained on more data (25M dataset) with basic architecture.

Constraint Summary: T4 VRAM Memory Wall -> Prioritized advanced model architecture over larger datasets (1M > 25M) for convergence.

RESULTS

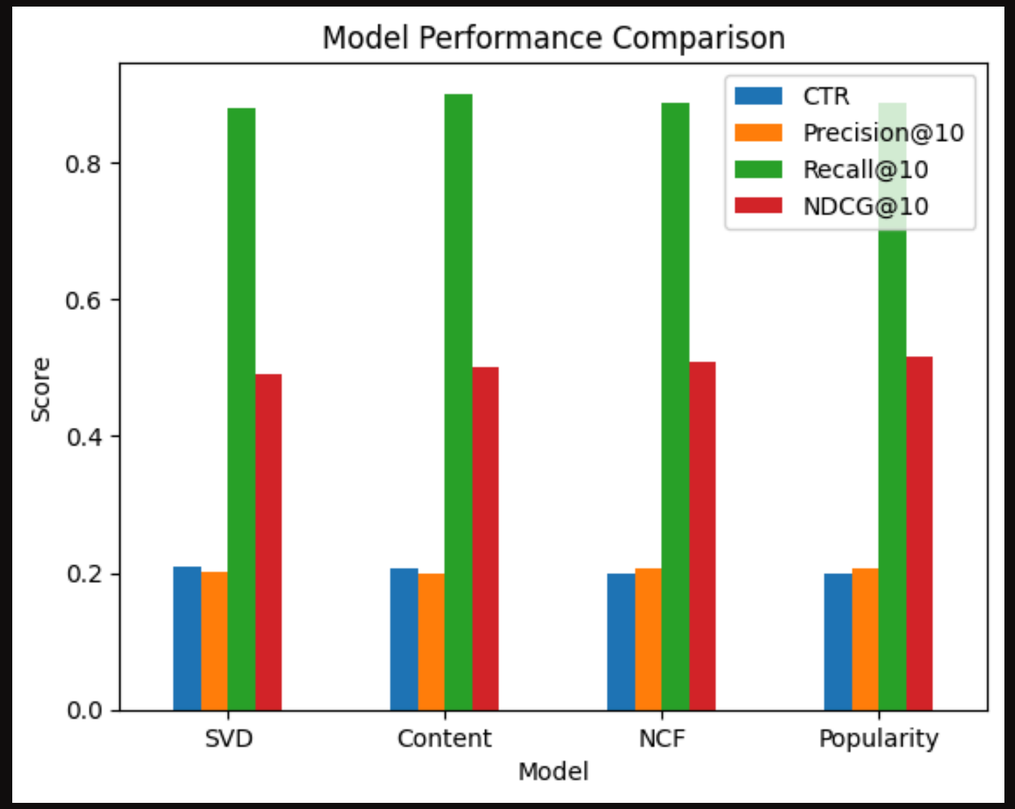
Model Comparison



Takeaway: CTR varies only slightly across models, suggesting limited practical improvement from increased model complexity

RESULTS

Model Comparison



Takeaway: No single model dominates all metrics. Performance differences are marginal.

RESULTS

Demo

Demo Video Link: https://www.youtube.com/watch?v=GKddulwY9lY

App link: https://personalizationengine.streamlit.app/

CONCLUSION

What We Learned

Data Quality and scale matter more than complexity

Evaluation Metrics

CONCLUSION

Future Work

BERT4Rec Sequential Recommendation

Extends the system using sequence-aware models like BERT4Rec to capture user behaviour over time

Thank You for Listening

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